MD KAMRUZZAMAN

Python Backend Developer | FastAPI | PostgreSQL | Docker | CI/CD

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GitHub | LinkedIn | Portfolio

Professional Summary

Python Backend Developer with a B.Tech in Computer Science and Engineering (Machine Learning specialization). Designed, built, and deployed two backend and ML systems using FastAPI, PostgreSQL, Docker, CI/CD, MLflow, and Kubernetes. Skilled in RESTful API development, authentication, database design, automated testing, containerization, deployment, monitoring, and ML model serving. Focused on clean architecture, reliability, and test-driven development.

Skills

Languages: Python, SQL, Bash

Backend: FastAPI, RESTful APIs, Pydantic, SQLAlchemy, Alembic

Databases: PostgreSQL, MySQL

Authentication & Security: JWT, bcrypt, OAuth2

DevOps & Cloud: Docker, Docker Compose, Kubernetes, GitHub Actions, CI/CD Pipeline, Git, Linux, Render, Neon

Testing & Quality: pytest, FastAPI TestClient, Ruff, test coverage

ML & MLOps: scikit-learn, MLflow, NumPy, Pandas

Observability & Monitoring: Prometheus, Grafana, Evidently

Projects

Expense Tracker API Code | Live Demo

- Architected and deployed a RESTful API for personal expense tracking with JWT authentication, bcrypt password hashing, and user-scoped data isolation enforced at the query layer.
- Designed a normalized PostgreSQL schema using SQLAlchemy ORM and managed schema changes with Alembic database migrations.
- Implemented pagination, category and amount filtering, Pydantic request validation, structured error handling, and correct HTTP status codes.
- Achieved 91% test coverage with pytest across authentication flows, CRUD operations, authorization, and edge cases.
- Containerized the application with Docker and configured a GitHub Actions CI/CD pipeline for automated testing and Ruff linting.
- Deployed on Render with PostgreSQL on Neon, exposing a public Swagger/OpenAPI interface for API documentation.

ML Expense Platform Code | Live Demo

- Built an end-to-end ML serving platform that predicts expense categories from free-text descriptions.
- Developed a scikit-learn TF-IDF and Logistic Regression pipeline; tracked four MLflow experiments with regularization parameter sweeps.
- Built a FastAPI prediction endpoint with Pydantic validation and lazy model loading for efficient inference.
- Containerized the service with Docker and wrote a Kubernetes deployment with two replicas, liveness and readiness probes, and self-healing behavior.
- Instrumented Prometheus metrics for prediction counts by category and prediction latency, visualized through Grafana dashboards.
- Implemented Evidently-based data drift detection and validated it by injecting simulated production data (50% of columns flagged as drifted).

Education

Lovely Professional University, Punjab, India 2025

B.Tech in Computer Science and Engineering, Machine Learning Specialization

Certifications & Training

- Meta Back-End Developer — Coursera Verify
- Natural Language Processing Specialization — DeepLearning.AI, Coursera Verify
- Python Data Structures — University of Michigan, Coursera Verify